
VVU EARTH TECH

Administrator Specification — Operations Outreach Edition

FOR OUTREACH DISTRIBUTION, SCOPING & SALES FRAMEWORK STRATEGIES

Target Audience: System Administrators, Operations Teams, Municipal IT

CLASSIFICATION: Outreach Distribution — Scoping & Sales Framework Strategy

Content Scope: Technical, Operational, and Strategic — Excludes Legal & Financial

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1. System Overview & Deployment Architecture

The **VVU EARTH TECH** platform is a deterministic evidence runtime for infrastructure monitoring, municipal water network management, and autonomous decision-making. It combines an **Epistemic DAG Runtime** (with four primitives: Fact, Proof, Policy, Projection), a **Trust Runtime** (5-state AIR safety pipeline), and a **72-Hour Resilience Matrix** to deliver verifiable, replay-deterministic, append-only evidence for every decision the system makes.

Deployment Architecture

Component	Deployment	Resources	Scaling
Epistemic Runtime	Kubernetes pod (runtime.yaml)	2 CPU, 4GB RAM, 20GB disk	HPA: 2-10 replicas
NATS Streaming	Kubernetes pod (streaming.yaml)	1 CPU, 2GB RAM, 50GB disk	Single cluster
Worker Pipeline	Kubernetes pod	1 CPU, 2GB RAM	HPA: 2-8 replicas
API Gateway	Kubernetes pod	0.5 CPU, 1GB RAM	HPA: 2-4 replicas
Evidence Collector	Kubernetes pod (evidence.yaml)	0.5 CPU, 1GB RAM	Fixed 2 replicas
Monitoring Stack	Prometheus + Grafana	2 CPU, 4GB RAM	Fixed deployment
Outreach Service	Kubernetes pod (outreach.yaml)	0.25 CPU, 0.5GB RAM	Fixed 1 replica

2. Kubernetes Infrastructure

VVU-VAL-001 deploys on provider-agnostic k3s with 6 Kubernetes manifests in the validation/VVU-VAL-001/kubernetes/ directory. All manifests use the vvu-validation namespace.

Manifest	Purpose	Key Configuration
namespace.yaml	vvu-validation namespace	Labels: app=vvu, tier=validation
runtime.yaml	Epistemic Runtime deployment	Resource limits: 2CPU/4GB, Liveness probe, Readiness gate
monitoring.yaml	Prometheus + Grafana stack	ServiceMonitor, alert rules, dashboard CM
evidence.yaml	Evidence collection pods	SHA-256 verification, hourly bundle archival
streaming.yaml	NATS durable queue	Cluster: vvu-nats, durable subscriptions, replay mode
outreach.yaml	Staged outreach enforcement	3-stage gate engine, recipient registry

3. Monitoring Stack

The monitoring stack consists of Prometheus (metrics collection), Grafana (visualization), and the VVU Mission Control Dashboard (public scoreboard). Key metrics are defined in `scoreboard/metrics-schema.json`.

Metric	Type	Source	Alert Threshold
circuit_breaker	enum (NORMAL/DEGRADED/RECOVERING/FAIL-CLOSED)	Runtime API	FAIL-CLOSED → Critical alert
facts_accepted	counter	Acceptance Pipeline	Drop > 10% → Warning
facts_rejected	counter	Evidence Compiler	Rejection rate > 5% → Warning
queue_depth	gauge	NATS streaming	> 10000 → Warning, > 50000 → Critical
latency_p99_ms	gauge	API gateway	> 500ms → Warning, > 2000ms → Critical
cpu_pct / ram_pct	gauge	Kubernetes	> 80% → Warning, > 95% → Critical
mmr_root	string (SHA-256)	MMR append	Mismatch → Critical (replay divergent)
replay_checksum	string (SHA-256)	Replay engine	Must match fact_log_checksum
spoofed_payloads quarantined	counter	Security gate	Must equal spoofed_payloads_injected

4. Circuit Breaker Operations

Breaker	States	Transition Rules	Admin Action
AIR Safety (5-state)	NORMAL/WARNING/TRIPPED/RECOVERY/ESCALATED	Score-driven + hysteresis	Observe only; DO NOT manually transition
Municipal (3-state)	NORMAL/DEGRADED/FAIL-CLOSED	Infrastructure-driven	Manual FAIL-CLOSED only for documented P5/P7 recovery

■ **ADMIN RULE:** No manual Circuit Breaker transitions except documented P5/P7 recovery sequences. Manual transitions on other phases = Critical failure = overall FAIL.

5. Validation Suite Operations

VVU-VAL-001 is a 72-hour continuous validation with 6 failure-injection phases. The protocol is pre-registered (frozen before T=0) with independent observers.

Phase	Name	Hours	Gate	Objective	Severity
P1	Nominal Load	0–12h	Baseline	Establish baseline under normal telemetry traffic	Critical
P2	Telemetry Flood	12–24 h	Acceptance Capacity	Verify acceptance pipeline absorbs 100× flood	Major
P3	Network Chaos	24–36 h	HLC Ordering	Verify replay stays deterministic under packet loss	Critical
P4	Storage Pressure	36–48 h	Append-Only Integrity	Verify graceful degradation under disk fill	Critical
P5	Node Failure	48–60 h	Recovery	Verify pods restart and no Fact is lost	Major
P6	Security Injection	60–66 h	HF-001/002/005	Verify every spoofed/malformed payload rejected	Critical
P7	Partition + Recovery	66–72 h	LVL-17	Verify deterministic HLC merge after partition	Critical

Milestone Tracking

Milestone	Name	Trigger	Actions
M00	Pre-Registration Published	Before T=0	Manual
M12	Nominal Phase Complete	Hour 12	Evidence checkpoint
M24	Flood Phase Complete	Hour 24	Evidence checkpoint
M36	Network Chaos Complete	Hour 36	Evidence checkpoint
M48	Storage Pressure Complete	Hour 48	Evidence checkpoint
M60	Node Failure Complete	Hour 60	Evidence checkpoint
M66	Security Injection Complete	Hour 66	Evidence checkpoint
M71	Partition Recovery + HLC Merge	Hour 71	Evidence merge
M72	Final Evidence Package Published	Hour 72	Validation complete

6. Operator Runbook

Golden Rules (Non-Negotiable)

Rule	Rationale
No code changes	Frozen commit hash must remain the build under test for 72 hours
No configuration edits	Config changes could alter runtime behavior mid-run
No manual Fact Log edits	Fact Log is append-only and immutable — edit = Critical failure
No manual Circuit Breaker transitions (except documented)	CB must transition per state machine
Hardware replacement only	Physical node failures: replace, log, sign entry
All interventions logged	Every SSH, kubectl, hardware touch → append-only operator log
All interventions signed	Every log entry signed with Ed25519 operator key
No touching evidence bundles	Evidence bundles immutable — no write access

7. Phase-by-Phase Monitoring Guide

Phase	Hours	What to Watch	When to Act
P1 Nominal	0–12	CB stays NORMAL, MMR progresses	Only if CB goes FAIL-CLOSED
P2 Flood	12–24	Queue depth, CB may go DEGRADED	Only if CB goes FAIL-CLOSED
P3 Network Chaos	24–36	Replay status, latency	Only if replay goes DIVERGENT
P4 Storage	36–48	Disk usage, CB DEGRADED expected	Only if disk fills to 100%
P5 Node Failure	48–60	Pods restarting, CB recovery	Only if pod doesn't restart in 5 min
P6 Security	60–66	Rejected payloads, HF gates	Only if spoofed payload is ACCEPTED
P7 Partition	66–72	NATS queue, then HLC merge	Only if merge produces conflicts

8. Critical Failure Response

- **1. Do NOT attempt to fix it.** The run terminates immediately; the outcome is FAIL.
- **2. Log the failure** in the operator log with timestamp and description.
- **3. Notify** the VVU engineering lead and independent observers.
- **4. Preserve all evidence** — do not delete or modify any logs, bundles, or state.
- **5. File a postmortem** within 48 hours, published alongside the evidence package.

■ ***A Critical failure is itself valuable evidence. The logs, artifacts, and postmortem are published alongside any subsequent successful rerun.***

9. Outreach Staged Release Enforcement

Communications Policy: VVU EARTH TECH follows a staged disclosure framework: (1) Engineering progress is communicated only after verification — not before. (2) Outreach is structured in three stages: Evidence Publication → Personalized Outreach → General Social & Press. (3) Sensitive operational details are shared on a need-to-know basis. (4) Mass-blast communication is structurally impossible — the outreach engine enforces sequential stage gates.

Sta ge	Name	Trigger	Audiences	Cooldown
1	Evidence Publication	M72 + Validation Index $\geq$ 90.0 + zero Critical failures	Technical communities, researchers	0h
2	Personalized Outreach	Stage 1 complete + 24h elapsed + no SEV-1	Investors, municipalities	24h
3	General Social & Press	Stage 2 complete + 48h elapsed	Journalists, general public	48h

■ *Mass-blast communication is structurally impossible — the outreach engine enforces sequential stage gates.*

10. Cape Town Pilot Deployment

Cape Town serves as the alternative pilot municipality case study for the VVU EARTH TECH Hydro-Gateway deployment. Cape Town was selected following a governance review that identified operational risks at the originally proposed municipality. Cape Town's established water management infrastructure, progressive governance framework, and existing IoT deployment experience make it an ideal partner for validating municipal-grade infrastructure monitoring technology.

Deployment Requirements

- **Network:** Secure municipal network with IoT sensor connectivity.
- **Hardware:** Hydro-Gateway acoustic leak detection sensors at designated monitoring points.
- **Software:** Epistemic Runtime deployed on municipal Kubernetes cluster.
- **Monitoring:** Prometheus + Grafana + VVU Mission Control Dashboard.
- **Operations:** Trained municipal IT staff with operator runbook access.

11. Resource Acquisition & Partnership Strategy

7-Track Resource Acquisition & Partnership Strategy: **Track A — Municipal Partnerships:** Active outreach to Cape Town and progressive municipalities for pilot deployment and validation partnerships. *Status: Active Outreach* **Track B — University Research Partnerships:** Engagement with South African and international universities for collaborative research, validation observation, and academic publication. *Status: Strategy* **Track C — Industry Integration:** Partnership development with water infrastructure, IoT, and municipal technology companies for integration, supply chain, and co-development. *Status: Strategy* **Track D — Public Funding:** Application to national and provincial technology innovation funds, water sector grants, and digital infrastructure programmes. *Status: Strategy* **Track E — Private Investment:** Structured engagement with impact investors, technology venture capital, and ESG-aligned funds. *Status: Strategy* **Track F — Sponsorship & Equipment:** Outreach to hardware manufacturers, cloud providers, and technology sponsors for equipment, infrastructure, and in-kind support. *Status: Strategy* **Track G — Community & Open Source:** Building developer community, open-source contributors, and civic technology networks for grassroots validation and adoption. *Status: Strategy*

Execution Principle: VVU EARTH TECH advances through multiple independent pathways in parallel. Every activity is subject to formal governance review. Progress is measured by verified engineering deliverables, not announcements. Where a pathway encounters obstacles, alternative pathways continue without delay. This parallel execution model ensures continuous forward progress regardless of individual pathway outcomes.